WBC Abnormality Detection

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Abstract

Blood cancer is among the critical health concerns worldwide and is associated with both genetic and environmental factors. Early detection plays a vital role in reducing mortality rates and improving treatment success. This study focuses on improving blood cancer diagnosis using advanced deep learning techniques. Several convolutional neural network (CNN) models, including ResNet50, RegNet-X, AlexNet, ConvNeXt, EfficientNet, InceptionV3, Xception, and VGG19, were implemented and evaluated on the C-NMC dataset for the classification of Acute Lymphoblastic Leukemia and Hematogones.The dataset was preprocessed using resizing, normalization, and augmentation techniques such as rotation, horizontal flipping, and contrast enhancement to improve model generalization. All models were trained using transfer learning with consistent hyperparameters and evaluated using accuracy, recall, F1-score, and confusion matrices.Among the evaluated models, RegNet-X achieved the highest performance . ConvNeXt and InceptionV3 also demonstrated strong performance, while classical models such as AlexNet and VGG19 showed lower accuracy This work demonstrates the potential of deep learning models in assisting early blood cancer detection, contributing to improved diagnosis and reduced mortality rates.

Keywords Blood cancer, Deep learning, White blood cells, RegNet-X, ConvNeXt, CNN, ALL, HEM

1. Introduction

White blood cells (WBCs), also referred to as leukocytes, are fundamental components of the human immune system. Produced in the bone marrow, they serve as the body's primary defense against infections and foreign pathogens. Abnormalities in WBC morphology, count, or differentiation are strong clinical indicators of a wide range of hematological disorders, most notably leukemia, lymphoma, and myelodysplastic syndrome. The accurate detection and classification of these abnormalities are therefore of paramount clinical importance. [1]

Hematological malignancies arising from abnormal WBC proliferation are classified as blood cancers, and they collectively represent a critical public health challenge. According to established literature, leukemia alone accounts for over 60,000 new diagnoses annually in the United States, with significantly higher global incidence when lymphoma and myeloma are included. Despite advances in treatment, late-stage diagnosis continues to be a major contributor to poor patient prognosis and elevated mortality rates. [2]

Traditional diagnostic approaches for WBC abnormalities rely on manual examination of peripheral blood smears or bone marrow aspirates by trained hematologists. While effective, this process is inherently labor-intensive, time-consuming, and subject to interobserver variability. The advent of digital pathology and automated image analysis has created an opportunity to mitigate these limitations through the deployment of machine learning and, more recently, deep learning algorithms capable of processing and classifying cellular images with high accuracy and consistency. [3]

Deep learning, in particular convolutional neural networks (CNNs), has demonstrated remarkable performance in medical image analysis tasks including segmentation, detection, and classification of histopathological images. These architectures can automatically extract hierarchical feature representations from raw image data, eliminating the need for manual feature engineering and enabling end-to-end learning pipelines. Several studies have reported CNN-based models achieving human-level or super-human performance on specific diagnostic tasks, including the classification of acute lymphoblastic leukemia (ALL) from microscopic blood smear images. [4]

The present work is motivated by the need for a robust, accurate, and computationally efficient automated system for WBC abnormality detection.

A systematic evaluation of multiple deep learning architectures for WBC abnormality classification from microscopic images.

Integration of stain-normalization-aware preprocessing and data augmentation strategies to improve model robustness and generalization.

A rigorous comparative analysis of model performance using standard evaluation metrics including accuracy, precision, recall, and F1 score.

Clinical relevance assessment of the best-performing model for potential deployment in computer-aided diagnosis (CAD) systems.

Compared multiple CNN architectures (AlexNet, VGG19, ResNet50, EfficientNet, ConvNeXt, InceptionV3, Xception, RegNet-X) for blood cancer classification.
Achieved high performance, with **RegNet-X** reaching **97.9%** accuracy, outperforming other models.
Applied advanced techniques such as transfer learning, data augmentation, TTA, and threshold tuning.
Provided comprehensive evaluation using accuracy, recall, F1-score, and confusion matrices

2. Related Works

2.1 Traditional Image Processing Methods

- Early approaches relied on classical image processing and hand-crafted feature extraction methods.
- Rahadi et al .examined classical image processing techniques for simultaneous detection of red blood cells (RBCs) and white blood cells from microscopic images, employing morphological operations and thresholding. While their accuracy was limited compared to deep learning approaches, the study provides a useful baseline and highlights the importance of preprocessing quality in any WBC analysis pipeline. [3]
- Ilyas et al. proposed a leukemia classification approach based on gene expression profiles using a linear programming-based computational technique. The method focuses on selecting the most relevant genes and optimizing classification boundaries to distinguish between different leukemia types. Their results showed improved classification accuracy compared to traditional statistical methods, highlighting the potential of optimization-based approaches in medical data analysis, although the method depends heavily on data quality and preprocessing. [4]
- Saikia and Devi. proposed a WBC classification approach based on Gray Level Cooccurrence Matrix (GLCM) features combined with Zero Phase Component Analysis (ZPCA) for dimensionality reduction. This hybrid feature-extraction method demonstrated competitive classification performance, particularly for distinguishing between lymphocytes and monocytes, though it was outperformed by end-to-end CNN approaches on larger datasets. [5]

2.2 Deep Learning Approaches

- The automated analysis of blood cell images for the purpose of detecting hematological abnormalities has attracted considerable research attention over the past two decades. More recent works have increasingly adopted deep learning frameworks.

- Lian et al. introduced a hybrid framework combining convolutional neural networks with generative adversarial networks (GANs) for acute leukemia prediction and classification. The GAN component was used to augment the training data, effectively addressing the problem of class imbalance that is frequently encountered in medical imaging datasets, and the combined approach demonstrated significant improvements in classification sensitivity for rare leukemia subtypes. [7]
- Asar and Ragab proposed a computer-aided diagnosis (CAD) system integrating deep learning with the Falcon Optimization Algorithm (FOA) for hyperparameter tuning.

Their system achieved state-of-the-art results on leukemia classification benchmarks, and the study highlighted the importance of systematic hyperparameter optimization in maximizing model performance. [8]

- Saeed et al. developed DeepLeukNet, a CNN-based model specifically adapted for microscopy images of acute lymphoblastic leukemia (ALL). The architecture employed transfer learning and fine-tuning strategies on the C-NMC dataset, demonstrating that domain-adapted pre-trained models can achieve high classification accuracy even with limited training data. [9]
- Raina et al. conducted a systematic review focused specifically on acute leukemia detection using deep learning techniques, covering literature from multiple international databases. Their findings confirmed the dominance of CNN-based architectures for this task, while also identifying critical challenges related to data imbalance, interpretability, and the generalizability of models trained on limited institutional datasets. [10]
- Han et al. proposed a one-stage and lightweight CNN architecture incorporating attention mechanisms specifically designed for WBC detection in microscopic images. Their model achieved competitive detection performance while significantly reducing computational overhead, making it suitable for deployment on resource-constrained hardware in clinical environments. [11]

2.3 Research Gap

Collectively, the reviewed literature establishes deep learning particularly CNN-based architectures as the dominant paradigm for WBC abnormality detection, while also revealing persistent challenges in dataset diversity, model interpretability, and clinical translation that this work aims to address.

2.4 Blood cancer definition

Some causes of leukemia include radiation exposure, family history, and exposure to certain chemicals. Most commonly, leukemia can be categorized based on the rate at which it progresses, and the type of cell involved. In many cases, the disease’s progression will initially classify leukemia into either one of two categories: acute leukemia and chronic leukemia.

2.5 Types of leukemia

They are mainly categorized depending upon the type of WBCs which is being affected. According to this category, there are two classes. First one is the Lymphocytic leukemia, through which lymphoid cells or lymphocytes, from which lymphatic tissues are formed gets affected. The second one is the Myelogenous leukemia which affects the myeloid cells, responsible for forming the RBCs, WBCs, and platelets in the blood. The second type of leukemia classification is according to the speed at which it affects the human body; for instance, acute leukemia. In acute leukemia, the cancerous cells are immature blasts. These cells grow and develop very fast and cannot perform their normal functions. Patients suffering from acute leukemia need vigorous and fast treatments.

Figure 1 explains the types of leukemia. Aside from those categorizations, leukemia has a multitude of subtypes. These subtypes include [16]:

- Acute lymphocytic leukemia, mostly occurring among children
- Acute myelogenous leukemia, common in adults and may affect children also
- Chronic lymphocytic leukemia is the most common chronic leukemia in adults
- Chronic myelogenous leukemia (CML), It could go undetected for many weeks, even months. By the time it is, the leukemia could have reached that gradual quick spreading phase [17].

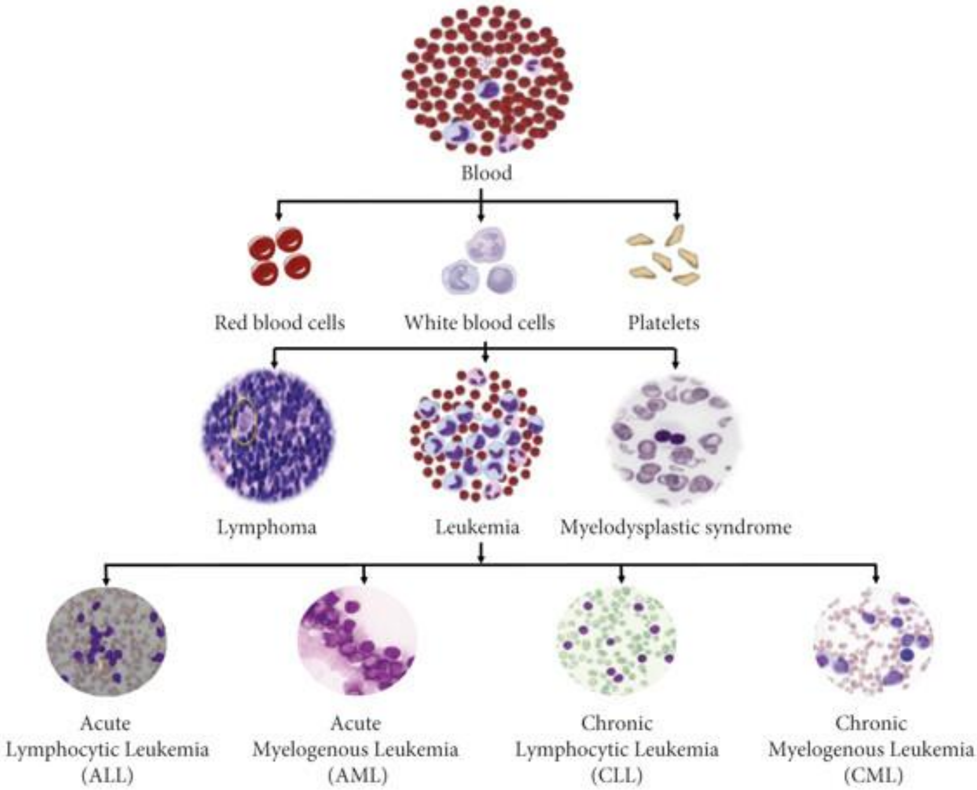

Fig. 1. Leukemia types [16].

Ref.	Techniques	Dataset	Accuracy	key contribution
[4]	EfficientNet	BCCD Dataset (Blood Cell Count and Detection Dataset)	92%	This paper uses deep learning to automatically detect and classify blood cells in leukemia patients to improve diagnosis accuracy and reduce manual work.
[4]	InceptionV3	BCCD Dataset (Blood Cell Count and Detection Dataset)	93%	The paper enhances blood cell detection and classification in leukemia using deep learning models, improving accuracy and supporting automated diagnosis.
[18]	ResNet-50	CellaVision dataset Raabin-WBC Kaggle WBC dataset	92.7%	The paper uses deep learning to classify and segment white blood cells, improving accuracy and providing detailed cell-level analysis.
[18]	Xception	CellaVision dataset Raabin-WBC Kaggle WBC dataset	90.37%	The paper applies deep learning to both classify and segment white blood cells, enabling more accurate and detailed analysis of blood cell images.
[19]	RegNetX	C-NMC 2019 Leukemia Dataset (CNMC Dataset)	96.4%	The paper proposes and compares multiple deep learning models for blood cancer classification, achieving very high accuracy and demonstrating the effectiveness of CNNs in medical diagnosis.

Table 1. List of Various Research Studies

3. Methodology

The methodology section gives detailed description of tool, techniques and steps taken while Building &Training ResNetRS50 and RegNetX016 deep learning model for cancer detection. The study includes the sub segments:

3.1 Dataset Collection and Description

- The dataset used in this study is the C-NMC (Cancer vs Normal Mitotic Cells) dataset, which contains microscopic images of blood cells classified into two categories:
    - ALL (Acute Lymphoblastic Leukemia) – cancerous cells
    - HEM (Hematogones) – normal cells

3.2 Preprocessing steps

- The dataset was preprocessed by resizing all images to a common resolution, applying normalization, and performing data augmentation techniques to improve model generalization and robustness.

    - **Resizing:**
      All images 224 × 224 pixels for most deep learning models (ResNet50, ConvNeXt, RegNet-X, VGG19, EfficientNet, AlexNet).
      299 × 299 pixels for InceptionV3 and Xception, as required by their architecture design.

    - **Normalization:**
      Pixel values were normalized using ImageNet mean and standard deviation to ensure compatibility with pretrained models and stabilize training.

    - **Denoising:**
      Although the dataset is relatively clean, normalization and resizing implicitly reduce noise effects and standardize input distribution for better feature extraction.

    - **Data Augmentation:**
      techniques were applied to increase dataset diversity and reduce overfitting.

    - **Rotation:**
      Random rotations were applied to images to simulate variations in cell orientation under microscopic imaging.

    - **Horizontal Flipping:**
      Images were flipped horizontally to generate additional realistic variations and improve model robustness.

    - **Normalization Consistency:**
      All transformations were applied consistently across training and validation datasets to ensure fair evaluation.

    - **Data Splitting:**
      The dataset was divided into training, validation, and testing sets using stratified sampling to preserve class distribution.
      (80% training, 20% testing)

3.3 Proposed Model

- Multiple deep learning architectures were used to evaluate performance
- The highest accuracy was achieved by the RegNet-X and ConvNeXt architecture.
- These models were selected due to their strong performance in image classification tasks.

3.4 Training Strategy

- **Optimizer:** AdamW
  Learning Rate: 1e-5
  Batch Size: 64
  Epochs: 40
  Loss Function: CrossEntropyLos
  Test Time Augmentation (TTA): Applied for final predictions
  Threshold Tuning: Used to optimize classification performance

- The training strategy was designed to ensure stable convergence and optimal performance across all models with consistent hyperparameters applied for fair comparison.

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4. Experiments and Results

4.1 Experimental Setup

- **Hardware:**
  - The experiments were conducted on a system equipped with a GPU Rx 5700xt
  - i5 14400f processor and 16gb ram to accelerate deep learning training
- **Software:**
  - The implementation was carried out using Python programming language with supporting libraries for deep learning and data processing.
- **Framework:**
  - The models were developed using PyTorch, with additional libraries such as Torchvision for pretrained models, NumPy for numerical operations, and Matplotlib/Seaborn for visualization.
- **Development Environment:**
  - The experiments were performed using google colab , enabling access to GPU resources and efficient model training.

4.2 Evaluation Metrics

- **Evaluation Metrics:**
  - Model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

4.3 Results and Analysis

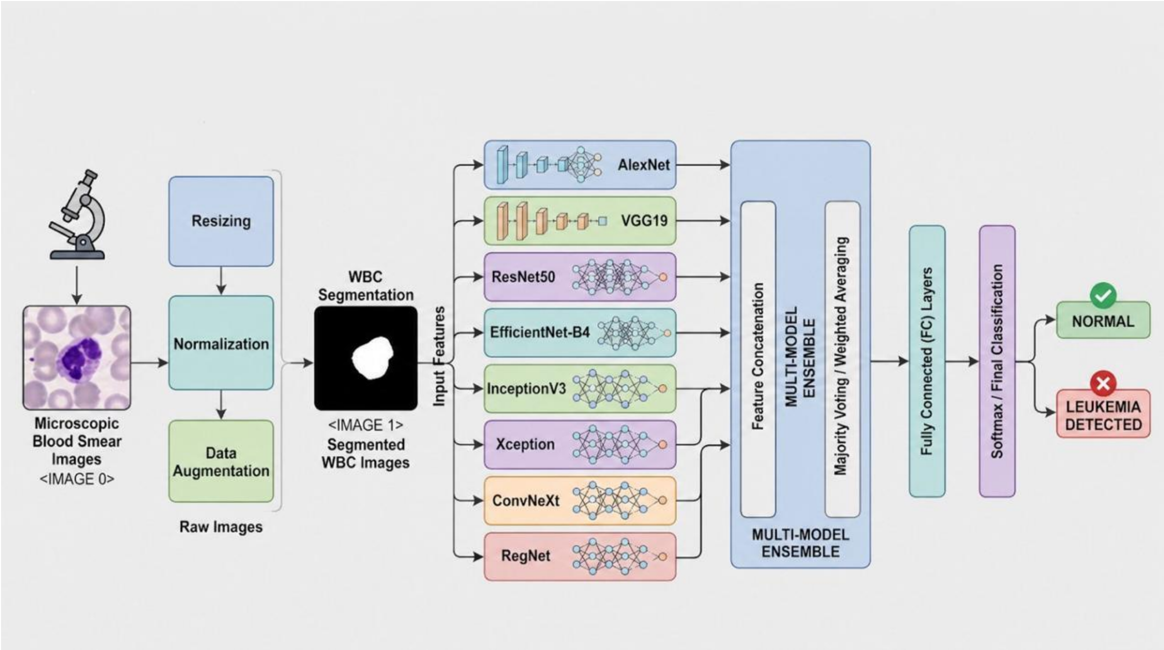

Fig. 2. Blood Cancer (Leukemia) Prediction System Architecture

1. **AlexNet**

- AlexNet is one of the earliest deep learning architectures that significantly improved image classification performance. It introduced deep convolutional neural networks with large receptive fields and ReLU activation, making training faster and more effective.
- **Architecture:** The network consists of five convolutional layers followed by max-pooling layers and three fully connected layers. It starts with an 11×11 convolution, followed by 5×5 and multiple 3×3 convolutions, ending with dense layers for classification.
- **Accuracy:** AlexNet achieved a major breakthrough on ImageNet and laid the foundation for modern CNN architectures. **(90.71%)**
- Despite its simplicity, it has a large number of parameters **(~61M)**, making it computationally heavy.

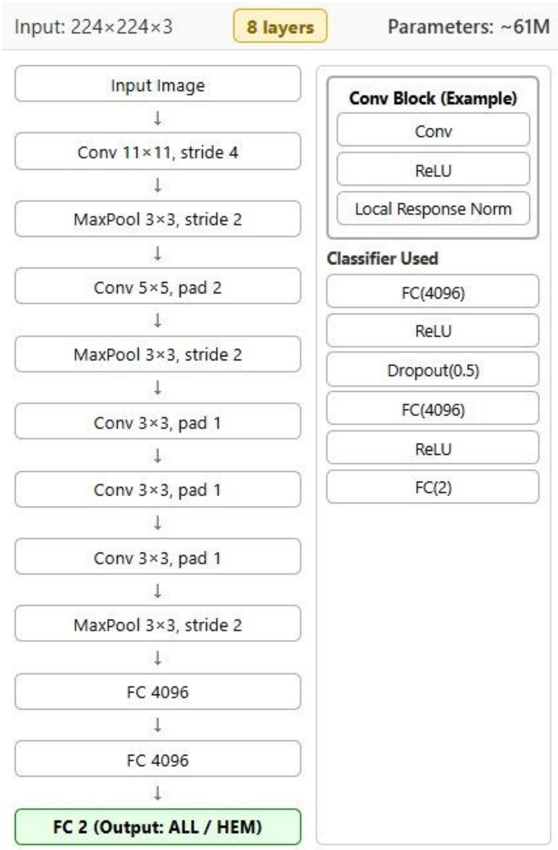

Fig. 3. AlexNet

2. **VGG19**

- VGG19 is a deep CNN architecture known for its simplicity and uniform design using small convolution filters.
- **Architecture:** It consists of five convolutional blocks with 2–4 layers each (3×3 convolutions), followed by max pooling. The network ends with three fully connected layers. It has **19 layers**.
- **Accuracy:** Provides good performance but with a very large number of parameters **(~143M)**. **(94.18%)**
- Computationally expensive but easy to understand and implement.

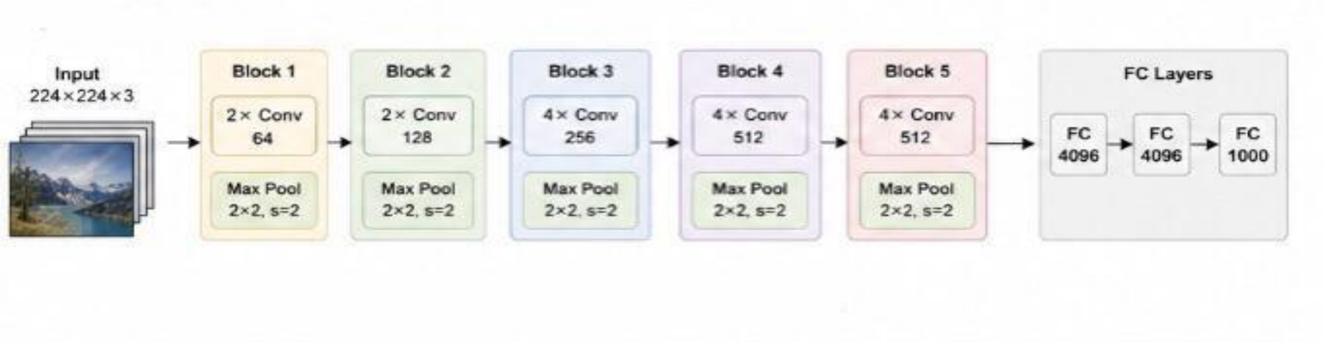

Fig. 4. VGG19 Model Architecture

3. **ResNet50**

- ResNet50 is a widely used deep neural network that introduced residual learning to solve the vanishing gradient problem.
- **Architecture:** It starts with a 7×7 convolution and max pooling, followed by four stages of bottleneck residual blocks (3, 4, 6, 3 blocks). Each block uses 1×1, 3×3, and 1×1 convolutions with skip connections. It has **50 layers**.
- **Accuracy:** ResNet50 achieves high accuracy **(~25M parameters)** and is widely adopted in many applications. **(96.60%)**
- Residual connections allow training very deep networks effectively.

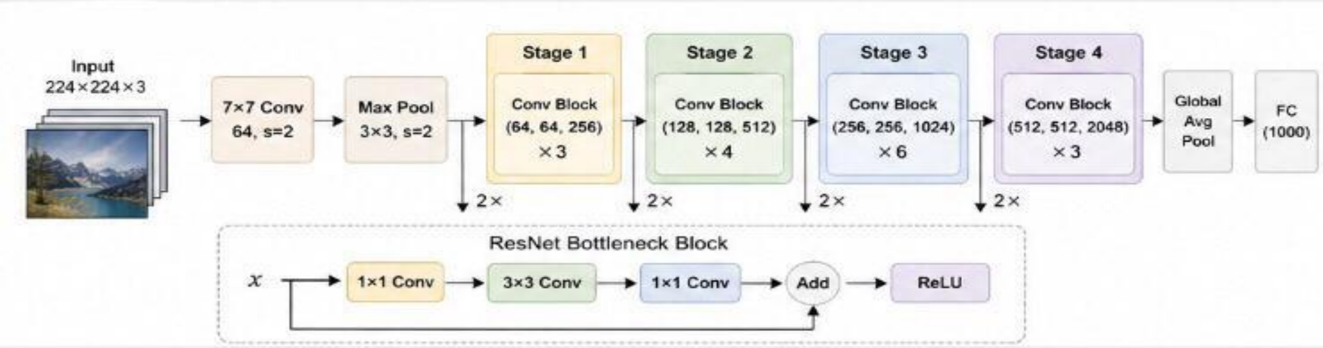

Fig. 5. ResNet50

4. **EfficientNet-B4**

   - EfficientNet-B4 is an advanced CNN architecture that uses compound scaling to balance depth, width, and resolution.
   - **Architecture:** It starts with a stem convolution followed by multiple MBConv blocks incorporating depthwise convolutions and squeeze-and-excitation mechanisms. The model ends with a head convolution, global pooling, and a fully connected layer. It has **~30 layers**.
   - **Accuracy:** EfficientNet models achieve high accuracy with fewer parameters **(~19M)** compared to traditional CNNs. **(92.50%)**
   - Known for its efficiency-performance trade-off, making it suitable for real-world applications.

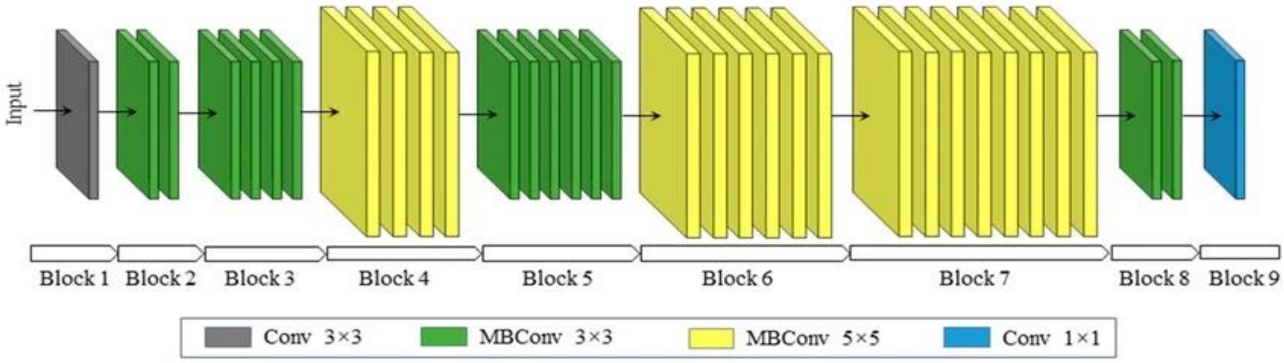

Fig. 6. EfficientNet-B4 Model Architecture

5. **InceptionV3**

   - InceptionV3 is a deep learning architecture designed to capture multi-scale features using parallel convolution operations.
   - **Architecture:** The model includes Stem layers followed by Inception-A, Reduction-A, Inception-B, Reduction-B, and Inception-C modules. Each module processes data using parallel convolution branches (1×1, 3×3, 5×5, and pooling) and concatenates the outputs. It has **48 layers**.
   - **Accuracy:** It provides strong classification performance with relatively low computational cost **(~24M parameters)**. **(96.81%)**
   - The model efficiently captures both local and global features.

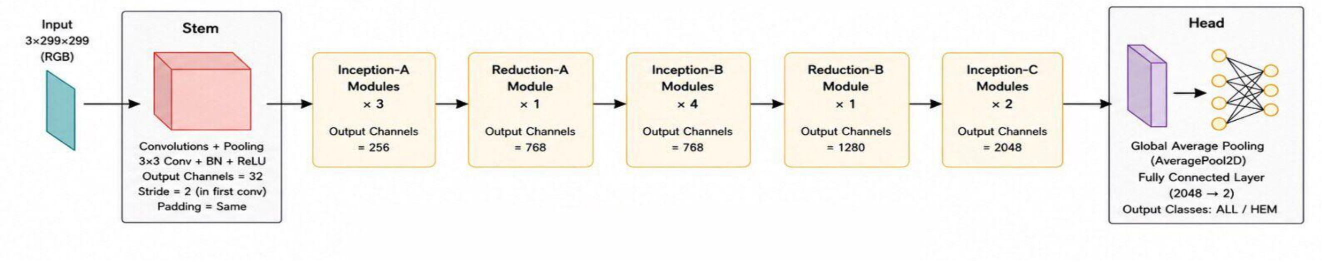

Fig. 7. InceptionV3 Model Architecture

6. **Xception**

- Xception is a deep CNN architecture based on depthwise separable convolutions, which improve efficiency and performance compared to standard convolutions.
- **Architecture:** It is divided into Entry Flow, Middle Flow (repeated multiple times), and Exit Flow. Each block uses depthwise separable convolutions followed by batch normalization and ReLU, with residual connections. It has **36 layers.**
- **Accuracy:** Xception achieves high performance with fewer parameters **(~23M)** due to efficient convolution operations. **(95.59%)**
- It is more efficient than traditional Inception models and widely used in medical image analysis.

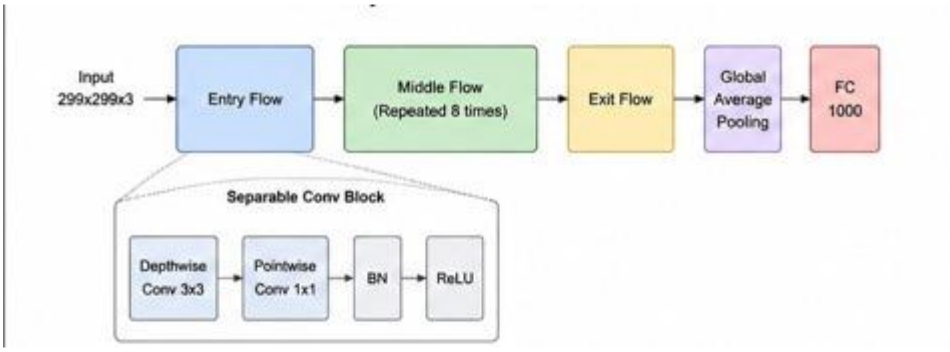

Fig. 8. Xception Model Architecture

7. **ConvNeXt**

- ConvNeXt is a modern CNN architecture inspired by Vision Transformers, combining convolutional efficiency with transformer-like design principles.

- **Architecture:** The model begins with a patchify convolution (4×4), followed by four ConvNeXt stages. Each block uses depthwise convolution (7×7), layer normalization, pointwise convolution, GELU activation, and residual connections. It has **~28 layers**.
- **Accuracy:** It achieves strong performance **(~28M parameters)** comparable to transformer-based models. **(96.90%)**
- Represents a new generation of CNNs with improved design strategies.

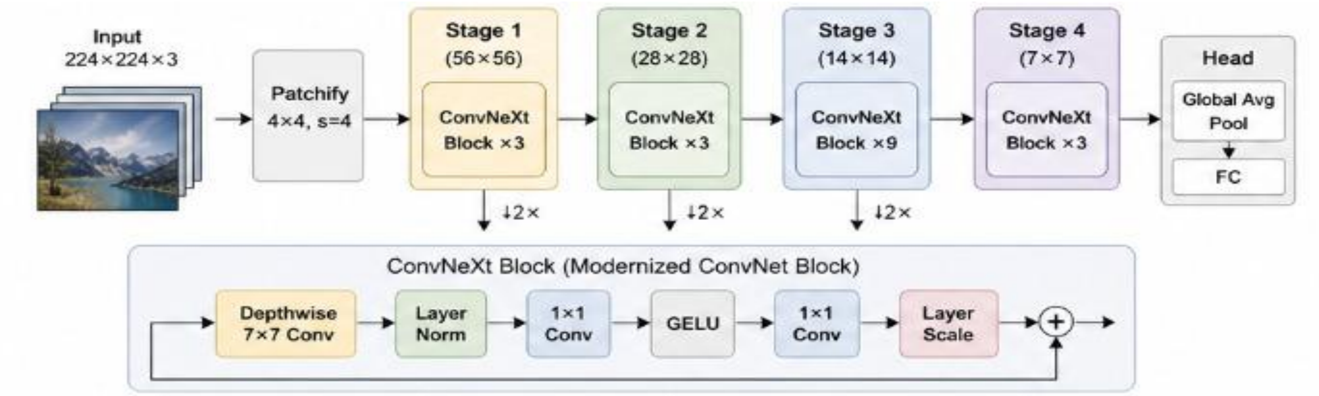

Fig. 9. ConvNeXt Model Architecture

8. **RegNet-X**

- RegNet-X is a scalable CNN architecture designed using a regular and systematic design approach.
- **Architecture:** The network includes a stem convolution followed by four RegNet stages using grouped convolutions, ending with global average pooling and a fully connected layer. It has **~40 layers**.

- Accuracy: Achieves strong performance **(~84M parameters)** with efficient resource utilization. **(97.94%)**
- Designed to balance accuracy and efficiency through structured design.

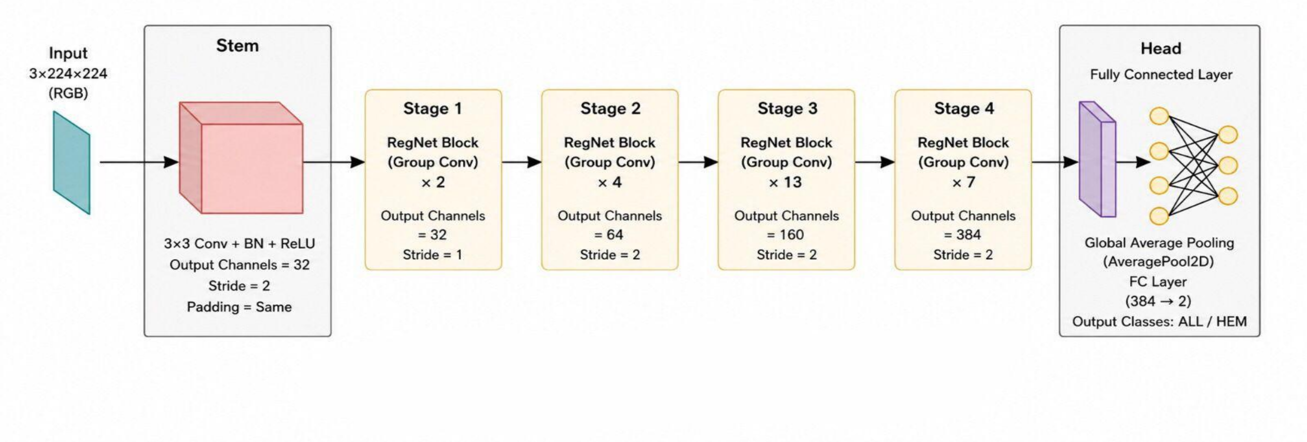

Fig. 10. RegNet-X Model Architecture

5. Discussion

5.1 Model Performance

The performance of the convNeXt and RegNetX models in wbc abnormality detections Both models demonstrated good accuracy, recall and F1 scores on the same size of dataset CNMC. However, it is important to delve into specific performance metrics to gain a deeper understanding of their capabilities (Table 2).

Figures 11 and 12 and Table 2 show, Accuracy was calculated for ResNet, RegNetX, AlexNet, EfficientNet, Inception_v3, Xception, VGG19 and Convnext, by using deep learning. It was found that the best percentage of Training set is **80%** and Testing Set **20%** at RegNetX algorithm because it gave us the highest Accuracy **(97.9%)**.

Algorithm	Accuracy	Recall	F1 Score
AlexNet	**90.7%**	**87.9%**	**89%**
ResNetRS50	**96.6%**	**96%**	**97%**
RegNetX	**97.9%**	**99%**	**98%**
Convnext	**96.9%**	**95%**	**96%**
EfficientNet	**92.5%**	**90%**	**90%**
Inception_v3	**96.8%**	**95.6%**	**96.28%**
Xception	**95.5%**	**94%**	**94.8%**
VGG19	**94%**	**92.5%**	**93.2%**

Table 2. Accuracy of models.

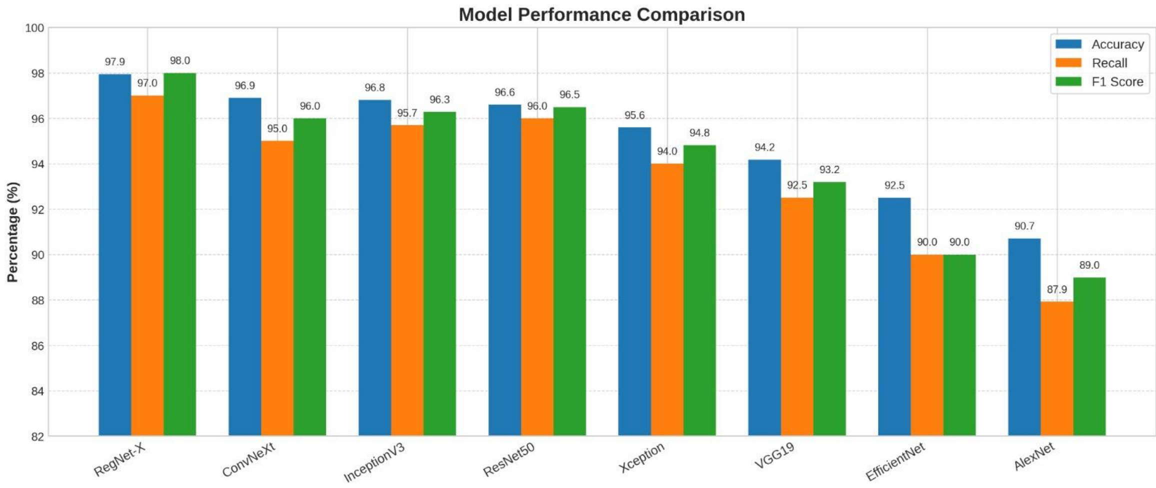

Fig. 11. Accuracy of models based on accuracy, Recall and F1 score

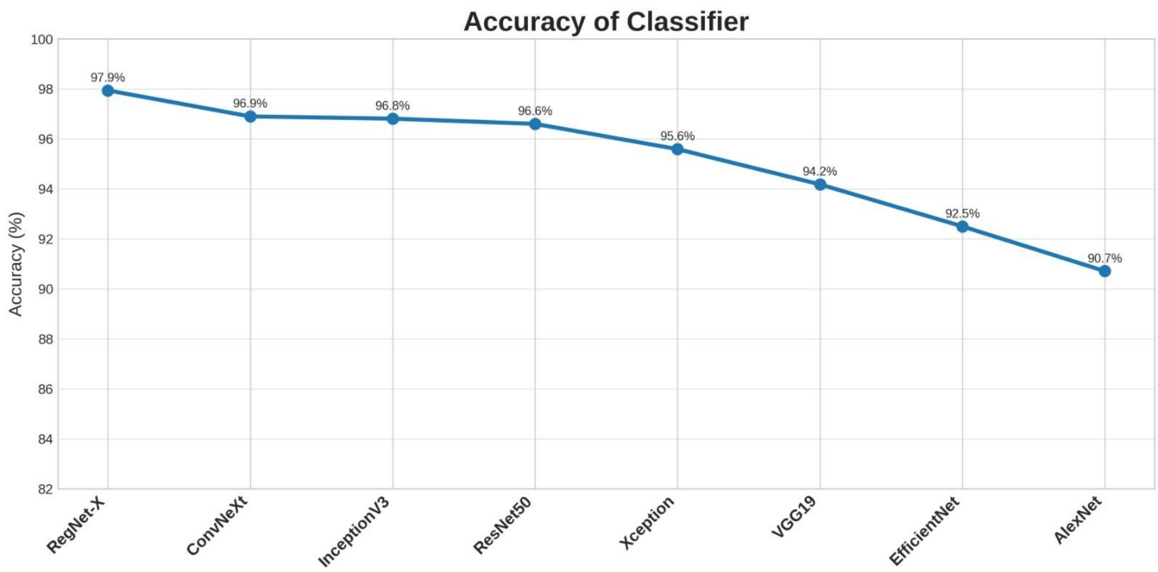

Fig. 12. Accuracy of Classifier

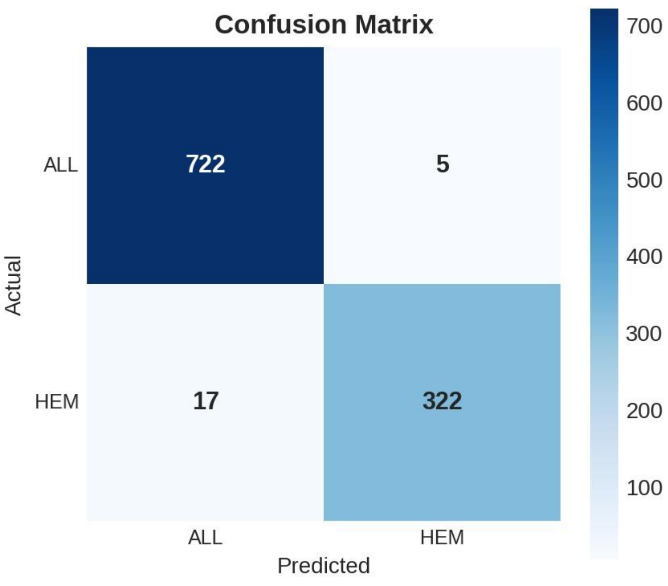

Fig. 13. Confusion matrix of RegNetX

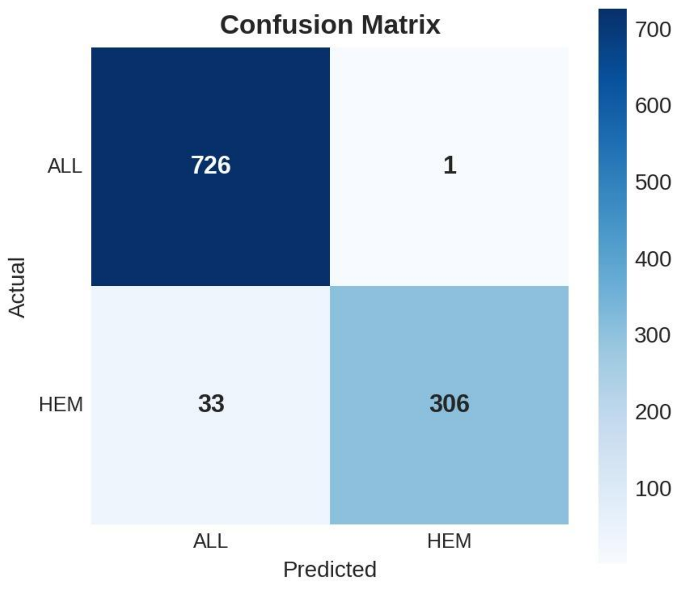

Fig. 14. Confusion matrix of ConvNeXt

5.2 Limitations

Limited dataset size:
Despite augmentation, the dataset is relatively small for deep CNNs, which can limit generalization. Large-capacity models like VGG19, RegNet-X may overfit without extensive data or stronger regularization.
Single-dataset evaluation:
Results are reported on one dataset without external validation, reducing confidence in real-world performance.
Computational requirements:
Top-performing models require significant GPU resources that's why we used cloud gpu from colab notebook
Limited interpretability:
Models act as black boxes so without explainability methods they may not be interpretable
Sensitivity to data quality:
Performance may degrade with poor image processing

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6. Conclusion and Future Work

6.1 Conclusion

the convNext and RegNetX have shown great clinical promise for blood cancer diagnosis automation, leading to better patient outcomes . However these need careful consideration of the ethical implications so that deployment is done in a manner that is equitable and beneficial.
Further research will be necessary to refine these models into real-world clinical applications. From the analysis, RegNetX scored an accuracy rate of 97.9% against 2.1% of error rates. This will therefore mean that RegNetX has a high possibility of becoming the helper for blood cancer diagnosis.

6.2 Future Work

- **Hyperparameter optimization:**
  - Advanced optimization techniques can be applied to further enhance model performance.
- **Clinical validation:**
  - Real-world clinical evaluation is necessary to assess model reliability and effectiveness in practical healthcare environments.
- **Interpretable AI:**
  - Integrating explainability methods and built in Ai will improve transparency and increase trust among clinicians.
- **Model enhancement:**
  - Future work may have ensembling and hybrid architectures and transform models to achieve higher accuracy.
- **Adverse event detection:**
  - Extending the model to detect disease progression or treatment-related changes can support better patient monitoring and classify the WBC types.
- **Global accessibility:**
  - to be accessible by the whole community and to take validation for it.

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